SF6024NT · THREE-CHUCK TUBE LASER CUTTING MACHINE (HORIZONTAL) · 1.5 – 6 KW

SF6024NT
TECHNICAL SPECIFICATION

THREE-CHUCK TUBE LASER CUTTING MACHINE (HORIZONTAL)









Ø0.8 – 9.1"	20 ft	Zero tail	0.8 G
ROUND TUBE	TUBE LENGTH	THREE CHUCKS	ACCELERATION

FULL TECHNICAL SPECIFICATION · SF6024NT · 1.5 – 6 KW

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- CSA-approved electrical panel
- 1-year warranty
- **7-day money-back guarantee** (in-stock machines only)
- Spare parts & technical support readily available in Canada
- **Order Today with Just 10% Down — 0% Down Financing Available**
- **Installation and training available**

SOLD & SERVICED IN CANADA

by NovoTech Machine Tools — Mississauga, ON

01 · MODEL · THREE-CHUCK TUBE LASER CUTTING MACHINE (HORIZONTAL)

SF6024NT

Versatile & efficient





Senfeng's campus in Jinan, China

1 SUPERIOR CUTTING STABILITY

2 POWERFUL DYNAMIC PERFORMANCE

3 THREE-CHUCK ZERO-TAIL SYSTEM

4 PNEUMATIC SELF-CENTERING CHUCK

**The picture is for reference; the actual appearance and size shall prevail.*

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02 · TECHNICAL PARAMETERS

TECHNICAL PARAMETERS · SF6024NT

NO.	ITEM	STANDARD 3 M UNLOADING SYSTEM	OPTIONAL 6 M UNLOADING SYSTEM	NOTE
1	**Tube length**	19.7 ft (6,000 mm)		
2	**Tube unloading length**	9.8 ft (3,000 mm)	19.7 ft (6,000 mm)	
3	**Tube diameter**	Ø0.8 – 9.1″ (Ø20 – 230 mm) bevel: Ø2 – 9.1″ (Ø50 – 230 mm)		National standard pipe
4	**Side length of square tube**	0.8 – 9.1″ (20 – 230 mm) bevel: 2 – 9.1″ (50 – 230 mm)		National standard pipe
5	**X / Y axis positioning accuracy**	±0.002″ (±0.05 mm)		
6	**X / Y axis repeated positioning accuracy**	±0.001″ (±0.03 mm)		
7	**Maximum feeding speed**	246 ft/min (75 m/min)		Processing speed, idle movement and other factors decrease as pipe weight increases.
8	**Maximum chuck speed**	100 rpm		
9	**Maximum acceleration**	0.8 G		
10	**Dimensions**	45.3 × 9.0 × 10.3 ft (13,800 × 2,750 × 3,150 mm)	54.0 × 9.0 × 10.3 ft (16,470 × 2,750 × 3,150 mm)	Subject to actual factory delivery
11	**Weight**	About 16,535 lb (7,500 kg)	About 18,629 lb (8,450 kg)	Subject to actual factory delivery
12	**Maximum load of chuck**	573 lb (260 kg)		Air pressure 44 – 102 psi (0.3 – 0.7 MPa)
13	**Phase**	Three-phase		
14	**Power supply rated voltage**	380 V — in Canada: 208 / 240 / 480 / 600 V 3-phase 60 Hz through the transformer NovoTech supplies		
15	**Frequency**	50 Hz		
16	**Power supply protection grade**	IP54		

1. The achievable accuracy of the workpiece depends to some extent on factors such as workpiece type, surface, tube specifications and straightness.

2. The technical parameters above are subject to change without notice; the final technical parameters are subject to the order agreement.

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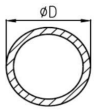
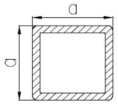
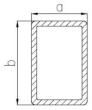
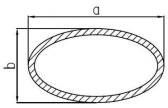
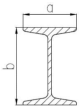
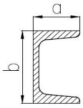
03 · TECHNICAL PARAMETERS

PRECAUTIONS · SF6024NT

1. Pipes being processed should be free from significant rust, as it can affect the cutting section quality.
2. Pipes should not be stored in the open air and should be placed flat, preferably with a surface oil coating or packaging. Ensure that the front end is even and the burr height at both ends is ≤0.20" (5 mm). Bundled pipes should have consistent lengths, with a deviation of less than 11.8" (300 mm).
3. For welded pipes, the external weld seam should be relatively flat, with a height of less than 0.012" (0.3 mm). The internal weld seam should have a height of less than 0.08" (2 mm).
4. The distortion and bending of the pipe should be lower than the tolerances specified in "GB/T 6728-2002 Dimensions, Shape, Weight, and Permissible Deviations of Cold-formed Hollow Sections for Structural Use."
5. The pipe should be straight, with a bending curvature of less than 0.012" per ft (1 mm/1 m; 3 mm over 6 m).
6. For the lengthwise torsion of the pipe, the total torsion should be less than 0.02 % of the total length.
7. The outer diameter tolerance of the profile should not exceed ±0.5 % of the outer diameter, with a minimum of 0.008" (0.2 mm)(according to the standard deviation level D4 specified in GB/T 17395-1998 for standardized outer diameter deviations).
8. When loading the pipes, ensure the safety of personnel and equipment. Bundle loading should be carried out using cranes operated by qualified and certified professionals. The crane should have both fast and slow speed functions, and slow speed should be used when approaching the loading platform.
9. Considering material errors, the positional error should be within IT12, and the shape and profile error should be within IT12.
10. The surface roughness of the pipe cross-section varies with the material and thickness.
11. Users must perform regular maintenance of the machine as instructed in the manual. Failure to do so may result in damage to the machine, and the manufacturer reserves the right to refuse repairs in such cases.
12. To ensure cutting effectiveness and minimize pipe deformation, the pipe wall thickness should meet the following criteria based on the pipe diameter:
    1. For pipe diameters greater than Ø2" (Ø50 mm): 1/40 ≤ wall thickness ≤ 1/10 of the pipe diameter.
    2. For pipe diameters smaller than Ø2" (Ø50 mm): 0.031" (0.8 mm) ≤ wall thickness ≤ 1/10 of the pipe diameter.
    3. Where the bending deflection exceeds the relevant standards, the cutting quality cannot be guaranteed.
13. Equipment temperature requirements: 32 – 104 °F (0 – 40 °C)(if the ambient temperature at the installation site exceeds this range, the buyer shall give written notice to the supplier at the time of purchase); the laser source should be placed in an environment of 64 – 77 °F (18 – 25 °C). If the ambient temperature at the installation site exceeds this range, air conditioning is recommended.
14. Equipment humidity requirements: ≤70 % (laser source only).
15. Cutting gas purity requirements: nitrogen ≥99.7 %; oxygen ≥99.5 %.

04 · CUTTING PARAMETERS

CLAMPING RANGE · SF6024NT

TYPE	CROSS-SECTION	CLAMPING RANGE
Round tube		Ø0.8 – 9.1″ (Ø20 – 230 mm) Ø20 – Ø230 mm
Square tube		a: 0.8 – 9.1″ (20 – 230 mm) □20 × 20 – □230 × 230 mm
Rectangular tube		a: 0.8 – 9.1″ (20 – 230 mm) · b: 0.8 – 9.1″ (20 – 230 mm) □20 × 20 – □230 × 230 mm
Elliptical tube		Please send the specific dimension drawing.
I-beam		min. a: 0.8″ (20 mm) · max. b: 7.9″ (200 mm) Standard configuration: GB No. 20
Channel steel		min. a: 0.8″ (20 mm) · max. b: 7.9″ (200 mm) Standard configuration: GB No. 20
Angle steel		Equal angle steel: a 0.8 – 4.3″ (20 – 110 mm) · max. b 6.1″ (155 mm) — GB No. 11 Unequal angle steel: a 0.8 – 3.5″ (20 – 90 mm) · b 0.8 – 5.5″ (20 – 140 mm) · max. c 6.5″ (166 mm) — GB No. 14/9

For bevel laser tube cutting, the minimum clamping size for pipes is 2″ (50 mm).

05 · CUTTING PARAMETERS · SF6024NT

CARBON STEEL

Cutting speed by laser source power in inches per minute, with Senfeng's metres per minute and the assist gas below each value.

Material	Wall thickness	1.5 kW	2 kW	3 kW	4 kW	6 kW
CARBON STEEL	**0.039″** 1 mm	394 – 512 10 – 13 · N_2 / air	591 – 787 15 – 20 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	1,181 – 1,772 30 – 45 · N_2 / air
	0.079″ 2 mm	157 – 197 4 – 5 · O_2	236 – 315 6 – 8 · N_2 / air	512 – 669 13 – 17 · N_2 / air	591 – 709 15 – 18 · N_2 / air	787 – 984 20 – 25 · N_2 / air
	0.118″ 3 mm	118 – 157 3 – 4 · O_2	118 – 157 3 – 4 · O_2	276 – 394 7 – 10 · N_2 / air	315 – 394 8 – 10 · N_2 / air	472 – 551 12 – 14 · N_2 / air
	0.157″ 4 mm	91 – 110 2.3 – 2.8 · O_2	98 – 118 2.5 – 3 · O_2	197 – 315 5 – 8 · N_2 / air	236 – 315 6 – 8 · N_2 / air	276 – 315 7 – 8 · N_2 / air
	0.197″ 5 mm	71 – 91 1.8 – 2.3 · O_2	87 – 102 2.2 – 2.6 · O_2	118 – 126 3 – 3.2 · O_2	118 – 126 3 – 3.2 · O_2	197 – 236 5 – 6 · N_2 / air
	0.236″ 6 mm	63 – 79 1.6 – 2.0 · O_2	71 – 79 1.8 – 2 · O_2	91 – 98 2.3 – 2.5 · O_2	98 – 110 2.5 – 2.8 · O_2	177 – 197 4.5 – 5 · N_2 / air
	0.315″ 8 mm	/	47 – 63 1.2 – 1.6 · O_2	71 – 87 1.8 – 2.2 · O_2	79 – 91 2 – 2.3 · O_2	87 – 98 2.2 – 2.5 · O_2
	0.394″ 10 mm	/	/	55 – 71 1.4 – 1.8 · O_2	63 – 79 1.6 – 2 · O_2	79 – 91 2.0 – 2.3 · O_2
	0.472″ 12 mm	/	/	39 – 55 1 – 1.4 · O_2	39 – 55 1 – 1.4 · O_2	75 – 83 1.9 – 2.1 · O_2
	0.551″ 14 mm	/	/	/	/	55 – 67 1.4 – 1.7 · O_2
	0.630″ 16 mm	/	/	/	/	47 – 55 1.2 – 1.4 · O_2
	0.709″ 18 mm	/	/	/	/	31 0.8 · O_2
	0.787″ 20 mm	/	/	/	/	/

1. For carbon steel pipes with a wall thickness over 0.197″ (5 mm) and a diameter or side length under 2″ (50 mm), oxygen is not recommended as the assist gas. Contact us for such pipe sizes.

2. There are two types of cutting heads: ① the standard cutting head cuts regular pipes such as square, round, elliptical and triangular pipes without concave angles; ② the PC cutting head makes a significant difference in cutting speed and quality on carbon steel pipes of 0.236″ (6 mm) and thicker — the speed is reduced by 20 – 40 %, but it can process pipes of any shape.

3. Parameters are for reference only. The values above are theoretical; the actual cutting speed may vary with factors such as pipe dimensions and equipment speed.

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06 · CUTTING PARAMETERS · SF6024NT

STAINLESS STEEL · ALUMINUM ALLOY

Cutting speed by laser source power in inches per minute, with Senfeng's metres per minute and the assist gas below each value.

Material	Wall thickness	1.5 kW	2 kW	3 kW	4 kW	6 kW
STAINLESS STEEL	0.039″ 1 mm	591 – 787 15 – 20 · N_2 / air	591 – 787 15 – 20 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	1,181 – 1,772 30 – 45 · N_2 / air
	0.079″ 2 mm	276 – 394 7 – 10 · N_2 / air	315 – 551 8 – 14 · N_2 / air	512 – 669 13 – 17 · N_2 / air	591 – 709 15 – 18 · N_2 / air	787 – 984 20 – 25 · N_2 / air
	0.118″ 3 mm	177 – 217 4.5 – 5.5 · N_2 / air	236 – 276 6 – 7 · N_2 / air	236 – 394 6 – 10 · N_2 / air	315 – 472 8 – 12 · N_2 / air	472 – 709 12 – 18 · N_2 / air
	0.157″ 4 mm	79 – 98 2.0 – 2.5 · N_2 / air	110 – 138 2.8 – 3.5 · N_2 / air	197 – 236 5 – 6 · N_2 / air	236 – 276 6 – 7 · N_2 / air	394 – 472 10 – 12 · N_2 / air
	0.197″ 5 mm	55 – 71 1.4 – 1.8 · N_2 / air	59 – 98 1.5 – 2.5 · N_2 / air	118 – 157 3 – 4 · N_2 / air	157 – 177 4 – 4.5 · N_2 / air	276 – 315 7 – 8 · N_2 / air
	0.236″ 6 mm	28 – 35 0.7 – 0.9 · N_2 / air	39 – 59 1 – 1.5 · N_2 / air	91 – 118 2.3 – 3 · N_2 / air	118 – 138 3 – 3.5 · N_2 / air	177 – 197 4.5 – 5 · N_2 / air
	0.315″ 8 mm	/	/	39 – 59 1.0 – 1.5 · N_2 / air	59 – 71 1.5 – 1.8 · N_2 / air	98 – 138 2.5 – 3.5 · N_2 / air
	0.394″ 10 mm	/	/	31 – 39 0.8 – 1 · N_2 / air	39 – 47 1 – 1.2 · N_2 / air	47 – 79 1.2 – 2 · N_2 / air
	0.472″ 12 mm	/	/	/	/	39 – 47 1 – 1.2 · N_2 / air
	0.551″ 14 mm	/	/	/	/	31 – 39 0.8 – 1 · N_2 / air
	0.630″ 16 mm	/	/	/	/	20 – 24 0.5 – 0.6 · N_2 / air
ALUMINUM ALLOY	0.039″ 1 mm	394 – 512 10 – 13 · N_2 / air	591 – 787 15 – 20 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	787 – 984 20 – 25 · N_2 / air	1,181 – 1,772 30 – 45 · N_2 / air
	0.079″ 2 mm	157 – 197 4 – 5 · N_2 / air	315 – 512 8 – 13 · N_2 / air	512 – 669 13 – 17 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	787 – 984 20 – 25 · N_2 / air
	0.118″ 3 mm	59 – 98 1.5 – 2.5 · N_2 / air	157 – 177 4 – 4.5 · N_2 / air	236 – 315 6 – 8 · N_2 / air	591 – 709 15 – 18 · N_2 / air	551 – 630 14 – 16 · N_2 / air
	0.157″ 4 mm	39 – 51 1 – 1.3 · N_2 / air	98 – 118 2.5 – 3 · N_2 / air	157 – 197 4 – 5 · N_2 / air	315 – 472 8 – 12 · N_2 / air	315 – 394 8 – 10 · N_2 / air
	0.197″ 5 mm	/	59 – 79 1.5 – 2 · N_2 / air	98 – 138 2.5 – 3.5 · N_2 / air	236 – 276 6 – 7 · N_2 / air	197 – 236 5 – 6 · N_2 / air
	0.236″ 6 mm	/	/	79 – 91 2 – 2.3 · N_2 / air	157 – 177 4 – 4.5 · N_2 / air	138 – 157 3.5 – 4 · N_2 / air
	0.315″ 8 mm	/	/	31 – 51 0.8 – 1.3 · N_2 / air	98 – 118 2.5 – 3 · N_2 / air	59 – 79 1.5 – 2 · N_2 / air
	0.394″ 10 mm	/	/	/	39 – 51 1 – 1.3 · N_2 / air	39 – 47 1 – 1.2 · N_2 / air
	0.472″ 12 mm	/	/	/	/	24 – 28 0.6 – 0.7 · N_2 / air
	0.551″ 14 mm	/	/	/	/	16 – 24 0.4 – 0.6 · N_2 / air

1. For carbon steel pipes with a wall thickness over 0.197″ (5 mm) and a diameter or side length under 2″ (50 mm), oxygen is not recommended as the assist gas. Contact us for such pipe sizes.

2. There are two types of cutting heads: ① the standard cutting head cuts regular pipes such as square, round, elliptical and triangular pipes without concave angles; ② the PC cutting head makes a significant difference in cutting speed and quality on carbon steel pipes of 0.236″ (6 mm) and thicker — the speed is reduced by 20 – 40 %, but it can process pipes of any shape.

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07 · COST-BENEFIT ANALYSIS

COST-BENEFIT ANALYSIS · SF6024NT

If the cutting auxiliary gas is dried compressed air, the cost includes the air compressor's electricity, the machine's power consumption and the consumables (protective lenses, cutting nozzles).

Item		3 kW			4 kW			6 kW		
		Air	O_2	N_2	Air	O_2	N_2	Air	O_2	N_2
PEAK POWER CONSUMPTION	Laser source (kW)	10			16			16		
	Chiller (kW)	3			6			6		
	Air compressor (kW)	15	/	/	15	/	/	15	/	/
	Host standard 3 m unloading	23.9	23.9	23.9	23.9	23.9	23.9	23.9	23.9	23.9
	Host optional 6 m unloading	25.2	25.2	25.2	25.2	25.2	25.2	25.2	25.2	25.2
	Dust removal (kW)	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
	Semi-automatic loading (kW)	3	3	3	3	3	3	3	3	3
Consumables and gas consumption (CNY/h)		0.5	4.5	60.5	0.5	4.5	60.5	0.5	4.5	60.5
Total power (kW) standard 3 m unloading		56.4	41.4	41.4	65.4	50.4	50.4	65.4	50.4	50.4
Total power (kW) optional 6 m unloading		57.7	42.7	42.7	66.7	51.7	51.7	66.7	51.7	51.7
Total power consumption (kWh per hour) standard 3 m unloading		33.84	24.84	24.84	39.24	30.24	30.24	39.24	30.24	30.24
Total power consumption (kWh per hour) optional 6 m unloading		34.62	25.62	25.62	40.02	31.02	31.02	40.02	31.02	31.02
Total operating cost (RMB per hour, at 1 RMB/kWh) standard 3 m unloading		34.34	29.34	85.34	39.74	34.74	90.74	39.74	34.74	90.74
Total operating cost (RMB per hour, at 1 RMB/kWh) optional 6 m unloading		35.12	30.12	86.12	40.52	35.52	91.52	40.52	35.52	91.52

1. The electricity and gas prices are Senfeng's reference values in China (RMB) and may vary by region.

2. Auxiliary gas consumption varies with plate thickness.

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08 · CONFIGURATION LIST

CONFIGURATION LIST · SF6024NT

NO.	ITEM	QTY	BRAND
		LASER SOURCE	
1	Laser source	1	Maxphotonics
		LASER CUTTING HEAD	
1	Laser cutting head	1	BOCI (optional bevel cutting head)
		MACHINE TOOL · HOST	
1	Transmission system	13	SENFENG
2	Machine tool and accessories	1	SENFENG
3	Reducer	9	TECHMECH
4	Electrical and pneumatic systems	1	AVENTICS
5	AC servo motor and driver	13	INOVANCE
6	Water chiller	1	TONGFEI
		CNC SYSTEM	
1	CNC system	1	FSCUT 5000A
		AUTOMATION SYSTEM (OPTIONAL)	
1	Loading system	1	Fully automatic / simplified fully automatic / semi-automatic (three options available)
2	Unloading system	1	Standard: 10 ft (3 m) · optional: 20 ft (6 m)

1. This is Senfeng's optimized configuration. Changes in brand or configuration may cause irreversible effects.

2. The warranty period for the entire machine (excluding consumables, non-force majeure natural disasters, war, improper operation, and human damage) is 1 year.

09 · CORE COMPONENTS

SF6024NT LASER HEAD



Standard cutting head



Tapered cutting head

1. **Integrated cooling:** uniform heat dissipation across the whole module allows long continuous cutting and extends the lens life 1.5 to 2 times.
2. **High sealing:** integrated body design with a dust protection rating of IP68.
3. **Easy maintenance:** the drawer-style design allows a lens change within three minutes.
4. **Standard cutting head:** for conventional tubes such as square and round tubes, angle irons, channel steels and other non-recessed profiles.
5. **Tapered cutting head:** processes tubes of any shape, including recessed profiles such as H-beams and I-beams.

OPTIONAL BEVEL CUTTING HEAD



Bevel cutting head (optional)

1. Supports ±45° bevel cutting.
2. Right-angle cutting head.
3. Full optical temperature and gas leakage monitoring.
4. Anti-collision bracket that protects the fiber interface and cables.
5. High-speed servo collimation and focusing.

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09 · CORE COMPONENTS

SF6024NT LASER SOURCE



Fiber laser sources — standard and optional

LASER SOURCE

1. **Small fiber core:** better beam quality.
2. **Faster cutting speed:** enhanced cutting efficiency.
3. **Improved high-reflection resistance:** new optical components with a multi-layer anti-reflection design for efficient processing of various metals.
4. **Enhanced intelligence:** next-generation software with a smart app mode for real-time monitoring.
5. **High electrical-optical conversion:** over 40 % efficiency, reducing power consumption.
6. **Upgraded components:** new-generation pump sources offer greater reliability and stability, with improved beam quality and 30 – 40 % more brightness thanks to patented nonlinear suppression technology.
7. **Lightweight and compact:** small size and compact design for easier integration.
8. **Enhanced safety:** improved safety features.

LASER SOURCE — OPTIONAL

1. Smaller core diameter, better beam quality.
2. More efficient cutting of thin and medium-thick plates.
3. Next-generation optics with a multi-layer anti-reflection design for better high-reflectivity resistance.
4. Smarter system with real-time dynamic monitoring of the operating parameters.
5. Electro-optical conversion efficiency over 40 %.
6. Modulation frequency range upgraded to 1 – 20 kHz, adaptable to diverse processes.
7. Compact size and reduced weight for easier integration.
8. Improved sealing and reliability with an upgraded IP66 protection rating.

INCLUDED WITH EVERY NOVOTECH MACHINE

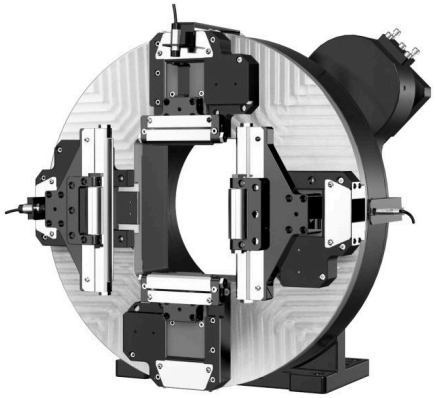
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- ✓ **Installation and training available**

10 · CORE COMPONENTS

SF6024NT CHUCK



Rear chuck



Middle chuck



Front chuck

1. Fully enclosed, integrated gear transmission structure for better dust protection and operational stability.
2. Two-front, one-rear chuck structure with high clamping precision, strong stability and a long service life.
3. The two front full-stroke chucks reduce jaw changes and boost production efficiency.
4. The front chuck has a square-opening design, allowing a wider clamping range of square and rectangular tubes.
5. The rear chuck has an extended barrel-type design that can insert into the front chuck, giving shorter tail material.

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11 · HOST MACHINE

SF6024NT HOST MACHINE





1 · Machine bed

1. The segmented bed has lifting rings on each section, reducing transportation costs.
2. The independently developed tube-welded bed offers better stability, a higher load capacity, stronger shock absorption and improved operational performance.
3. The bed is designed on structural dynamics principles and finite element variable analysis. After precision welding it undergoes stress-relief annealing, secondary aging treatment and high-precision machining on a large gantry milling machine — high rigidity and a long service life.

2 · Follow support

1. The servo-controlled system automatically aligns the tube center with the chuck center, ensuring constant surface contact with the tube.
2. The intelligent segmented layout keeps long tubes from sagging, ensuring high processing accuracy.



3 · Automatic unloading system

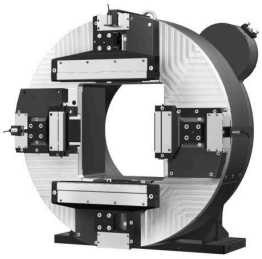
1. An automatic material separation system separates scrap and finished tubes without manual intervention.
2. The servo control system automatically aligns the tube's center with the chuck's center, preventing sagging under gravity and ensuring high processing accuracy.
3. After processing the plate flips automatically, so the finished tubes slide into the receiving rack.

12 · HOST MACHINE

THREE-CHUCK STRUCTURE



Rear chuck



Middle chuck



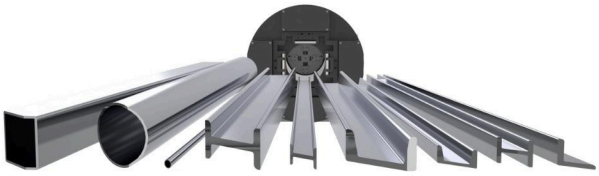
Front chuck

4 · THREE-CHUCK STRUCTURE

1. The standard configuration is a 1 rear + 2 front chuck structure, with automatic unloading of tubes up to 9.8 ft (3 m). Tubes longer than 9.8 ft (3 m) are unloaded by hand (which needs additional space), so workpieces of varying lengths can be processed.
2. It cuts with zero tail material, improving material utilization and saving costs.
3. All three chucks are movable. The rear chuck can pull and support the material, helping to correct workpiece deformation and improving accuracy and stability for very long or small workpieces.
4. Innovative "seven-axis material pulling" dynamic cutting, with the front hollow clamping chuck, makes the most of the raw material.

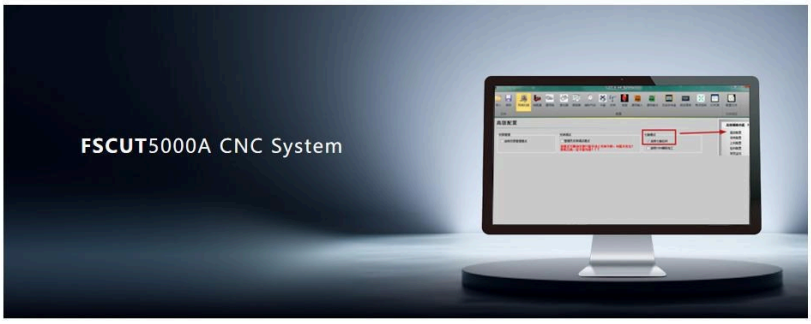
5 · POWERFUL TUBE PROCESSING CAPABILITY

1. **Size:** round tube Ø0.8 – 9.1" (Ø20 – 230 mm); square tube 0.8 × 0.8 – 9.1 × 9.1" (20 × 20 – 230 × 230 mm).
2. **Pipe type:** round, square, rectangular and elliptical tube, angle steel, channel steel, I-beam, H-beam, etc.
3. **Single tube load-bearing capacity:** 573 lb (260 kg).



13 · CNC SYSTEM

FSCUT5000A CNC SYSTEM



FSCUT5000A is a bus control system specially developed for pipe processing and suitable for four-chuck pipe cutting. It supports high-precision, high-efficiency cutting of square, round, runway-shaped and oval stretched pipes, as well as angle steels, channel steels, I-beams and other special-shaped steels. Its main performance features are as follows:

- Real-time compensation of the pipe center deviation reduces pipe clamping requirements and is more efficient; piercing accuracy is higher and cutting more accurate.
- Based on the bus real-time system and integrated follow-up control, pipe corner cutting is faster, corner following more stable and cutting quality more reliable.
- Supports the support-bracket follow-up function, ensuring the cutting quality of the whole pipe.
- Supports hollow chuck feeding and cutting, greatly shortening the total length of the bed, with a high degree of automation.
- Supports three-chuck pulling cutting, ensuring the cutting accuracy of long parts and zero-tail cutting.
- Supports automatic loading and unloading and cycle processing, and the standard automatic feeding action.
- Supports the chuck avoidance / cutting head avoidance function for processing shorter tails.
- Supports the groove (bevel) processing function (optional).
- Supports the chuck anti-burning function, which solves chuck burning caused by grooving.
- Supports special-shaped steel cutting — a more powerful function.
- Track accuracy 0.0008" (0.02 mm), positioning accuracy 0.00004" (0.001 mm), repeat positioning accuracy 0.0001" (0.003 mm).
- Supports real-time encoder feedback, with error measurement tools that help find the best motion parameters of the machine.

14 · CNC SYSTEM

TUBEPRO CUTTING SOFTWARE + TUBEST NESTING SOFTWARE



- Automatically supports editing the paths of special-shaped steel such as angle steel, channel steel, I-beam and H-beam for high-quality cutting.
- Centering function with real-time deviation compensation of the tube center.
- Built-in drawing function: fast drawing, and standard parts imported in batches.
- Common-edge nesting reduces the number of cuts — higher cutting efficiency and extreme material saving.
- A separate corner process can be set to improve the quality of pipe corner cutting.
- The Z axis also takes part in interpolation — 80 % control + 20 % follow-up, climbing and downhill — so the Z axis is no longer the weak link and corners are cut faster.
- Fast "frog jump": intelligent judgment of the lifting height on a face change shortens empty moves in all directions and improves cutting efficiency.
- Weld compensation automatically avoids the convex corners of vertical cutting, for interference-free splicing and easy intersecting welding.
- A simple interface, convenient operation, easy to learn, with a high degree of intelligence.
- Supports multiple formats such as IGS / SAT / JHB.

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15 · OPTIONAL CONFIGURATION

SF6024NT OPTIONAL UNITS

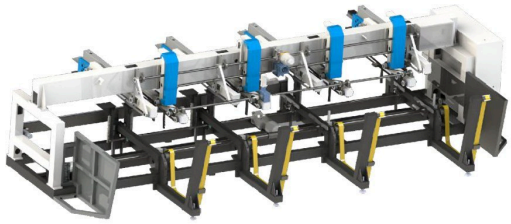
1 · SEMI-AUTOMATIC LOADING SYSTEM optional



Minimum loading tube length	13.1 ft (4,000 mm)
Maximum loading tube length	19.7 ft (6,000 mm)
Tube section side length (or diameter)	0.8 × 0.8 – 8.7 × 8.7″ (20 × 20 – 220 × 220 mm); Ø0.8 – 8.7″ (Ø20 – 220 mm)
Maximum single tube weight	529 lb (240 kg)
Tube type	Round, square and rectangular tube, angle steel, channel steel, I-beam, etc.

After placing tubes on the rack, arrange them neatly by hand; tubes beyond the specified size or weight are loaded by hand.

2 · FULLY AUTOMATIC LOADING SYSTEM optional



Loading tube length	11.5 – 19.7 ft (3,500 – 6,000 mm)
Tube section side length (or diameter)	0.8 × 0.8 – 9.4 × 9.1″ (20 × 20 – 240 × 230 mm); Ø0.8 – 9.1″ (Ø20 – 230 mm)
Maximum single tube weight	573 lb (260 kg)
Warehouse weight capacity	6,614 lb (3 t)
Tube type	Round tube, square tube

3 · SIMPLE AUTOMATIC LOADING SYSTEM optional

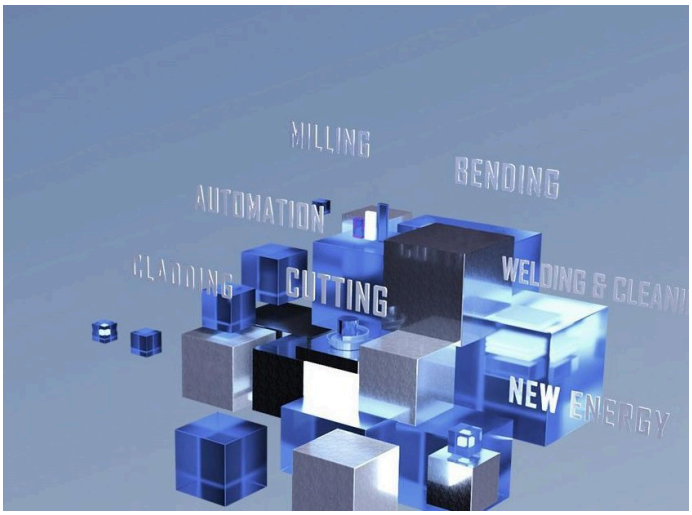


Loading tube length	11.5 – 19.7 ft (3,500 – 6,000 mm)
Tube section side length (or diameter)	0.8 × 0.8 – 9.1 × 9.1″ (20 × 20 – 230 × 230 mm); Ø0.8 – 9.1″ (Ø20 – 230 mm)
Maximum single tube weight	573 lb (260 kg)
Warehouse weight capacity	6,614 lb (3 t)
Tube type	Round tube, square tube

16 · AFTER-SALES SERVICE

PROVIDING GLOBAL USERS WITH AUTOMATED METAL FABRICATION SOLUTIONS

Senfeng provides automated metal fabrication solutions globally, integrating cutting, bending, welding, cladding, automation and new energy. With core technologies in laser cutting, welding and cladding, the company also offers intelligent manufacturing systems such as flexible laser processing lines and panel benders. These solutions serve industries including automotive, construction, energy and petrochemicals, and are used in over 100 countries.



03
INTERNATIONAL PCT PATENTS

577
CHINESE TECHNOLOGY PATENTS

Senfeng has developed key components such as laser generators, processing heads, the Feng Cloud system and CNC systems, used in cutting, welding, cladding and automation. The company offers a complete industry chain, including laser cutting, welding, cladding and cleaning machines, bending centers and flexible production lines. These solutions are used in sectors such as power towers, construction machinery, shipbuilding, bridges and aerospace, helping businesses accelerate production and reduce costs for greater economic benefit.

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17 · AFTER-SALES SERVICE

TRAINING · PACKAGING · INSTALLATION

TECHNICAL TRAINING TO ENHANCE CUSTOMER PRODUCTION EFFICIENCY

01 BEFORE EQUIPMENT DELIVERY

The buyer can arrange for 1 – 2 operators to attend a one-week training at the Senfeng factory. The dates are confirmed with Senfeng's customer service department.

02 DURING THE WARRANTY PERIOD

The buyer can apply for one more session of free training for 1 – 2 operators at Senfeng.

03 TRAINING

Laser principles, equipment structure, process explanation, equipment maintenance, laser safety protection, operating procedures and basic troubleshooting.

04 REQUIREMENT

Trainees must be mechanical, electrical or optical engineers and pass assessments on equipment operation, laser principles, safety and maintenance before starting work.

PACKAGING & TRANSPORTATION TO ENSURE EQUIPMENT QUALITY







01 PACKAGING

Standard packaging, suitable for long-distance transportation, moisture-proof, rust-proof and shock-resistant. Designed for full lifting, with marked lifting points and center of gravity.

02 TRANSPORTATION

Domestic transportation within China is fully handled by Senfeng, including freight and insurance.

03 PACKING LIST & CERTIFICATE

Each package includes a packing list and a certificate of conformity. The user manual and other documents are inside the box; the packing list is on the outside.

INSTALLATION PROFESSIONAL AND HIGH-QUALITY

01 INSTALLATION

NovoTech engineers, with Senfeng, install the equipment at the customer's site.

02 EQUIPMENT DEBUGGING

After installation, the equipment is debugged for the customer's use.

03 DOOR-TO-DOOR TRAINING

On-site training on equipment maintenance, safety, operating procedures and basic troubleshooting, with 7 days of guaranteed usage.

04 ON-SITE ACCEPTANCE

After the steps above, the engineer leaves the site only after the customer's acceptance (the customer can veto).

18 · AFTER-SALES SERVICE

PREMIUM CUSTOMIZATION

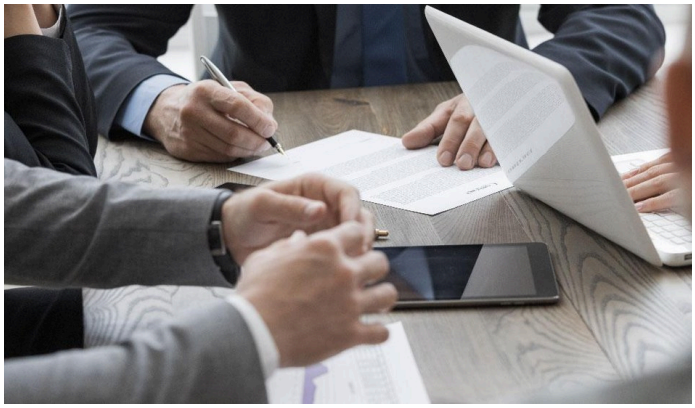
In the digital era, intelligent transformation in metal processing is essential. Building unmanned factories is key to upgrading, with customized automation solutions a top priority.



01 · Senfeng understands the client's industry status and specific production processes, identifies issues and assesses their needs.



02 · Through in-depth on-site communication, metal fabrication automation solutions are customized to the client's pain points and needs.



03 · A tailored solution, different from competitors' standardized models.



04 · A one-stop R&D and production service from design to final machine, with thorough quality checks until customer satisfaction and acceptance.

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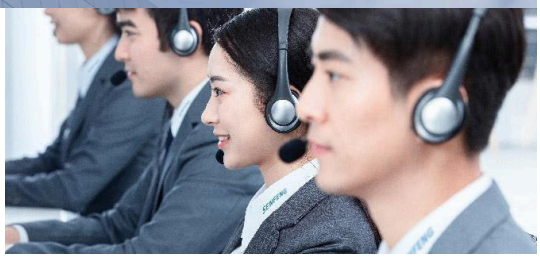
19 · AFTER-SALES SERVICE

FIVE-STAR FAST SERVICE



EFFICIENT

- Senfeng's repair hotline is available 24/7.
- A professional engineer responds to customer inquiries within 10 minutes and provides a repair plan within 1 hour.



PROFESSIONAL

- **Custom service:** service plans tailored to the customer's specific needs.
- **Service engineer certification:** each service engineer is rigorously trained and assessed before being certified to work.
- **Common issues training:** a manual of common issues for each equipment model, with certified engineers training customers.
- **Online guidance:** experienced senior engineers support customers by phone or video call.
- **Professional technical support:** the equipment is properly calibrated at the first installation, and similar issues are resolved in one go.





COMPREHENSIVE

- **Pre-service:** theoretical and hands-on operation training, common fault self-diagnosis training, quick repair guidance and usage reminders.
- **Regular service:** regular maintenance reminders, on-site services and periodic promotional activities.
- **Value-added service:** equipment software and hardware upgrades, financing lease services and extended warranty services.



20 · AFTER-SALES SERVICE

CUSTOMER TESTIMONIALS

Senfeng machines and customers around the world.



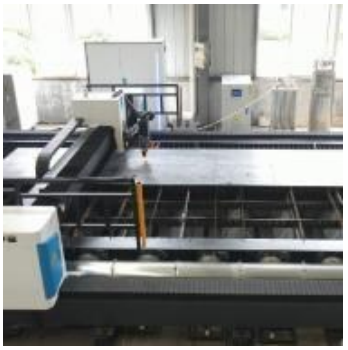
















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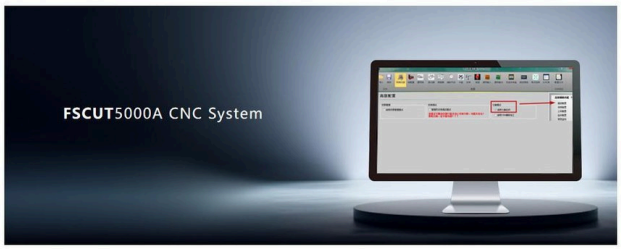
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Specifications subject to change without notice. Imperial values are converted from Senfeng's metric data and rounded: sheet thicknesses in decimal inches, cutting speeds in inches per minute. Model numbers are manufacturer designations. Photos may show optional equipment.